Plotting from Dataframes

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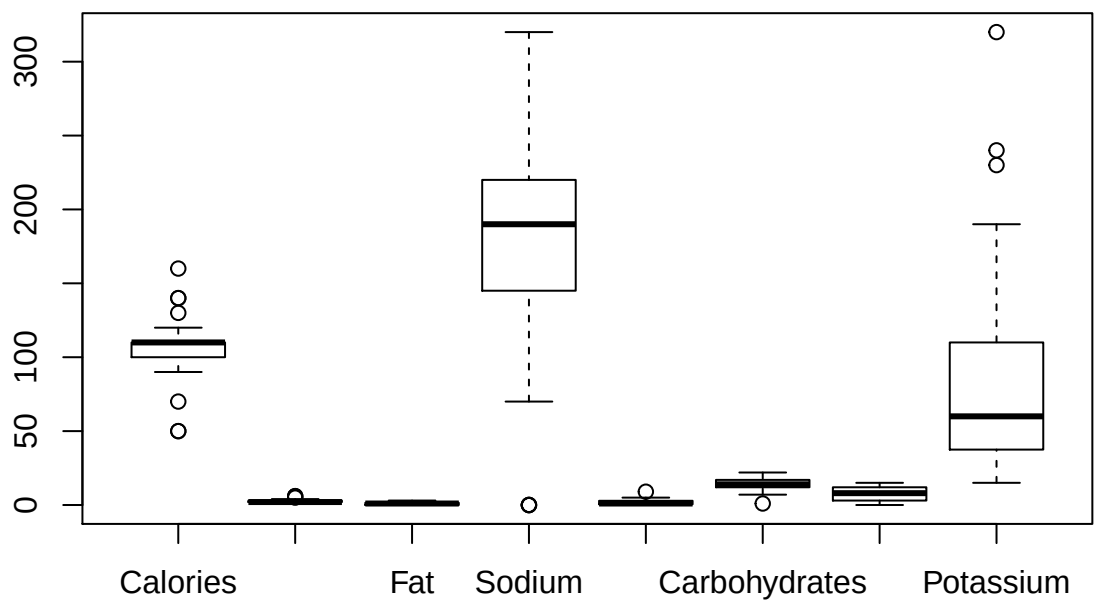
Intro to plotting

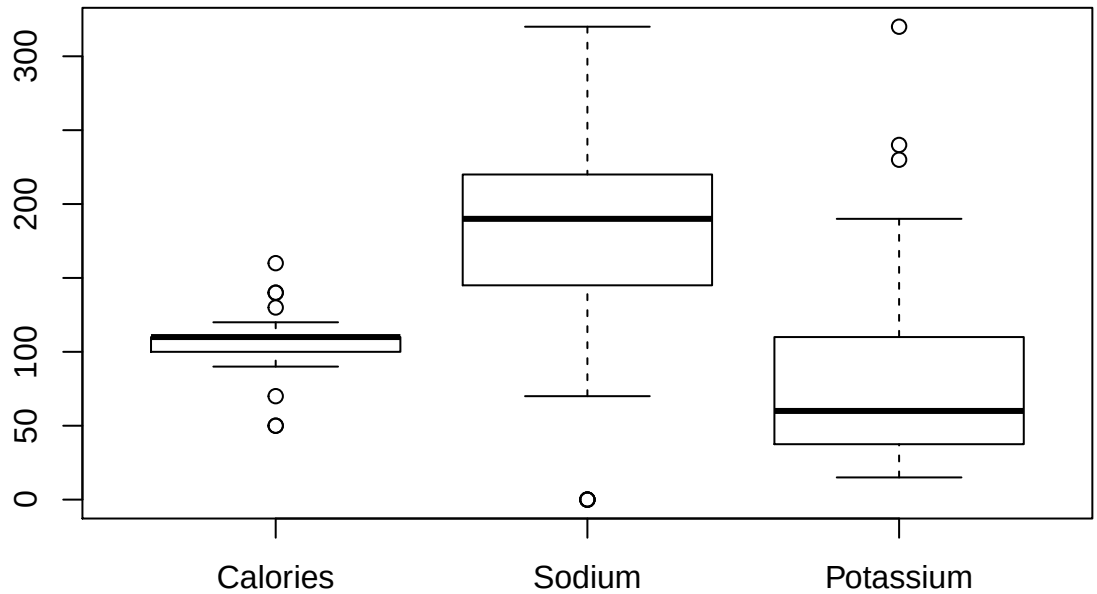
```
cereal <- read.csv('cereal.csv')
```

boxplot() Function

```
boxplot(cereal[,-1])
```

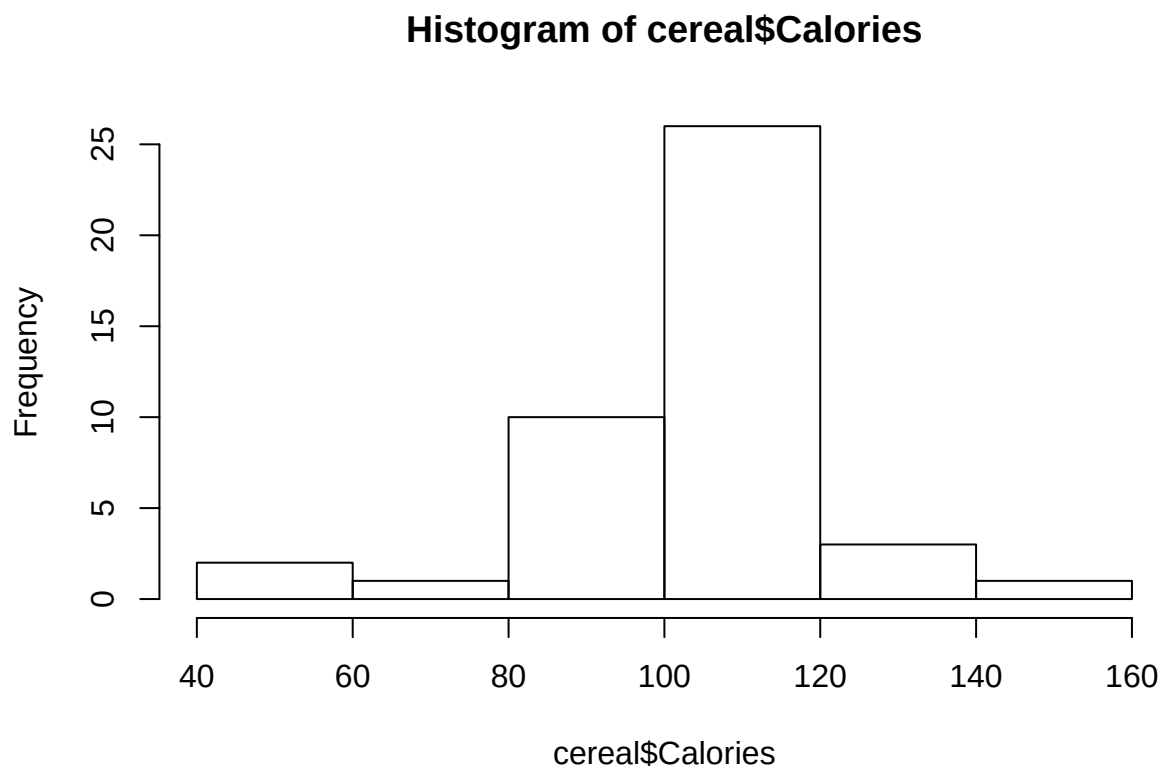


```r
boxplot(cereal[,c('Calories','Sodium', 'Potassium')])
```

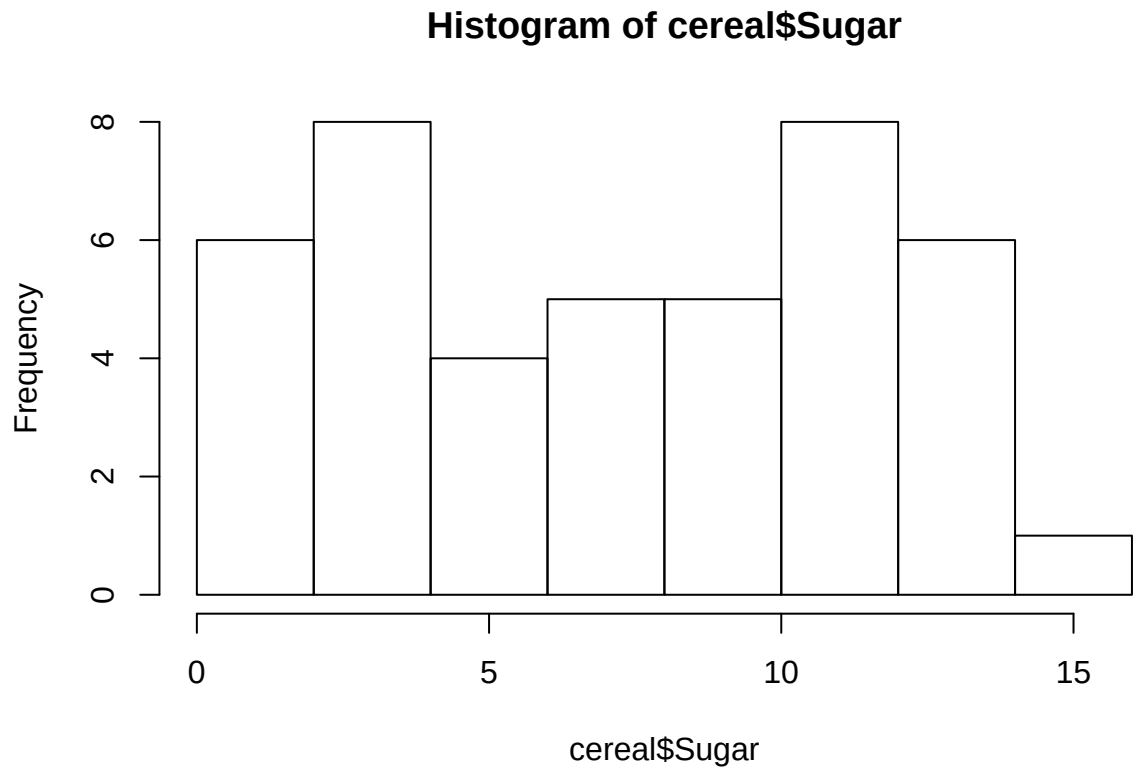


hist() Function

```r
hist(cereal$Calories)
```

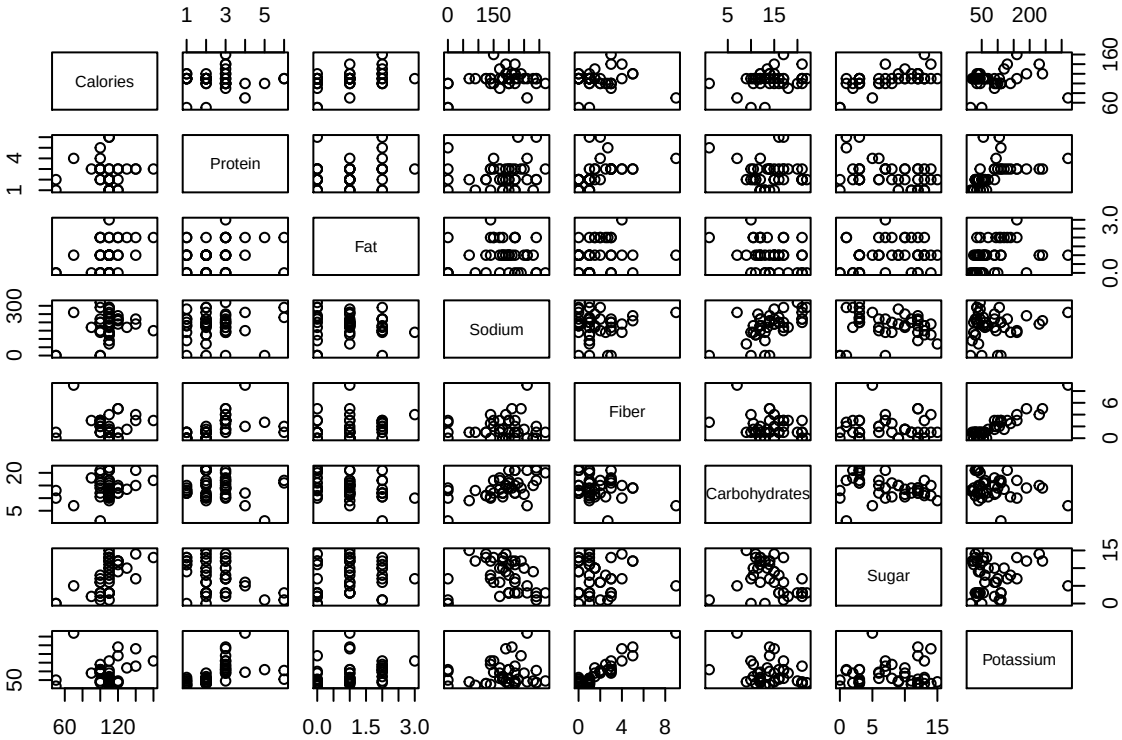


```r
hist(cereal$Sugar)
```



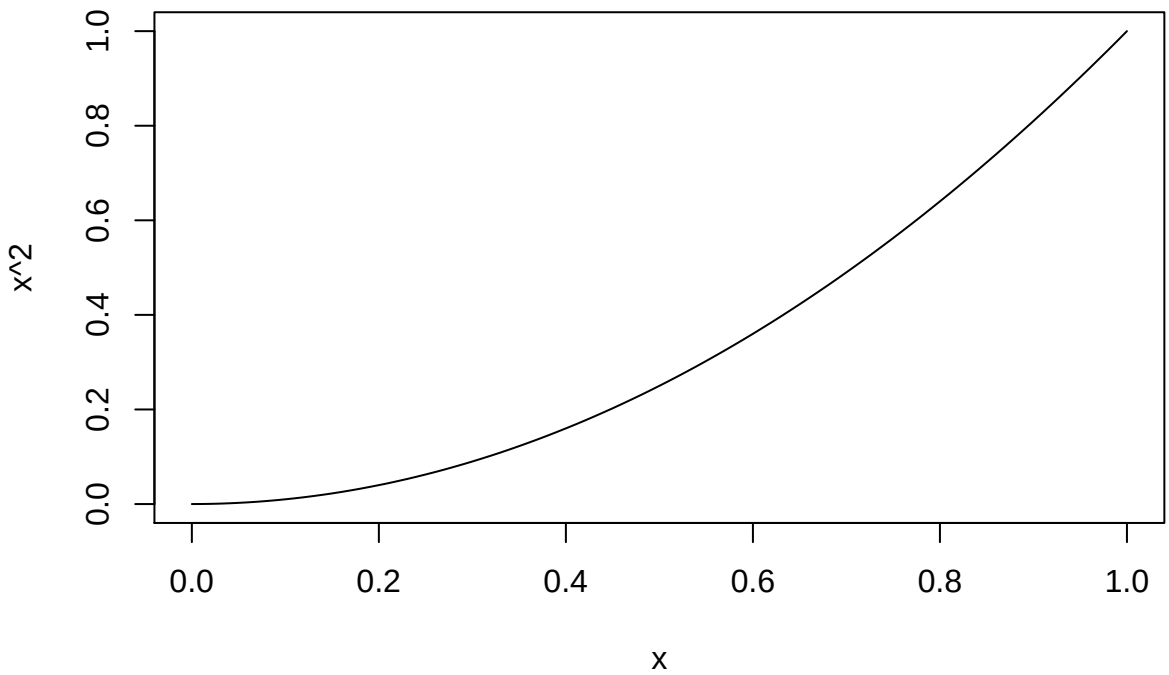
pairs() Function

```r
pairs(cereal[,-1])
```

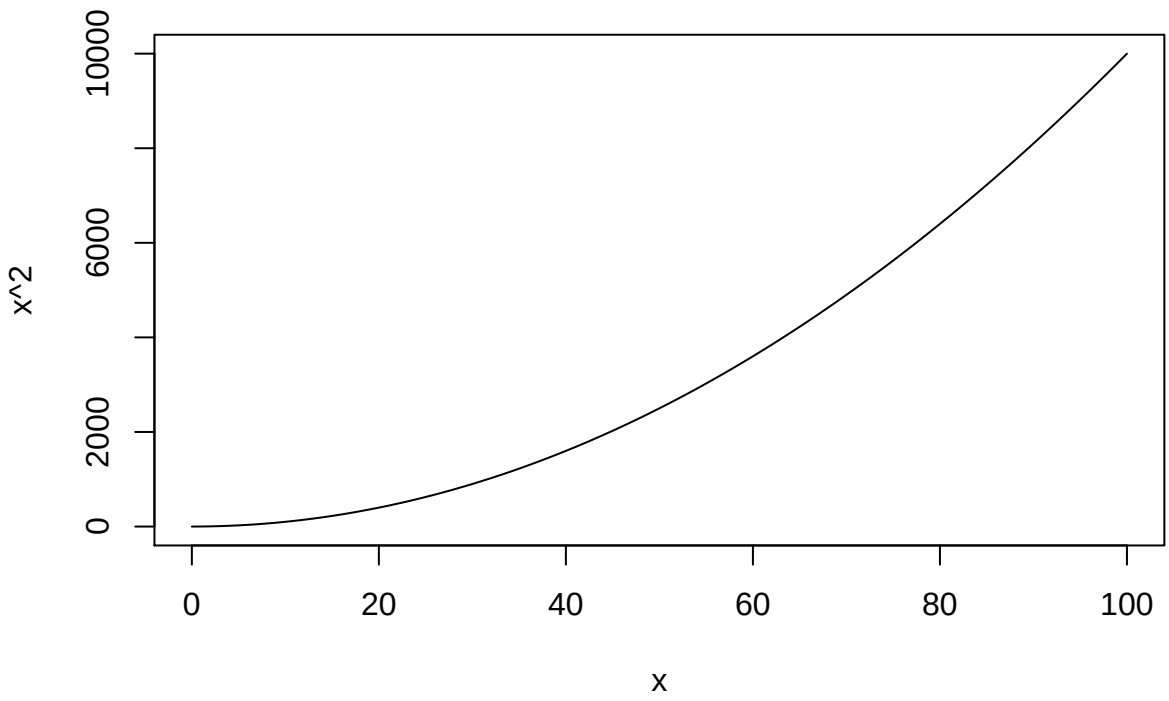


curve() Function

```r
curve(x^2)
```



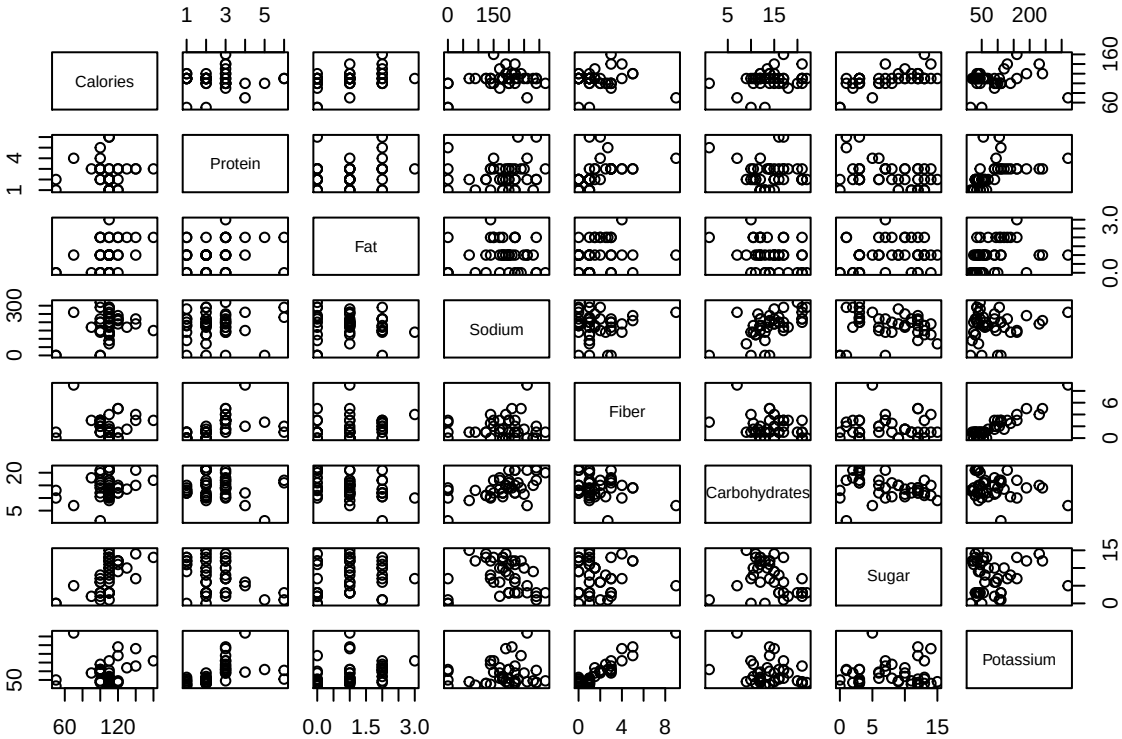
```r
curve(x^2, 0, 100)
```



plot() Function

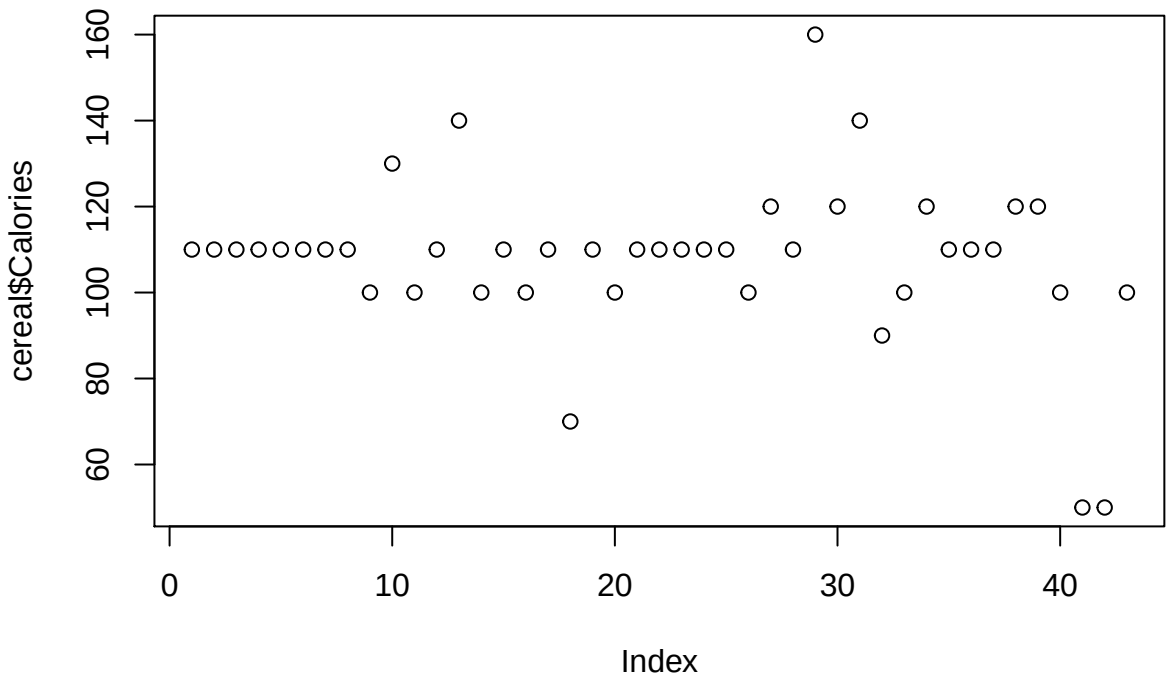
Running plot on whole dataset will be equivalent to pairs() Function

```r
plot(cereal[,-1])
```

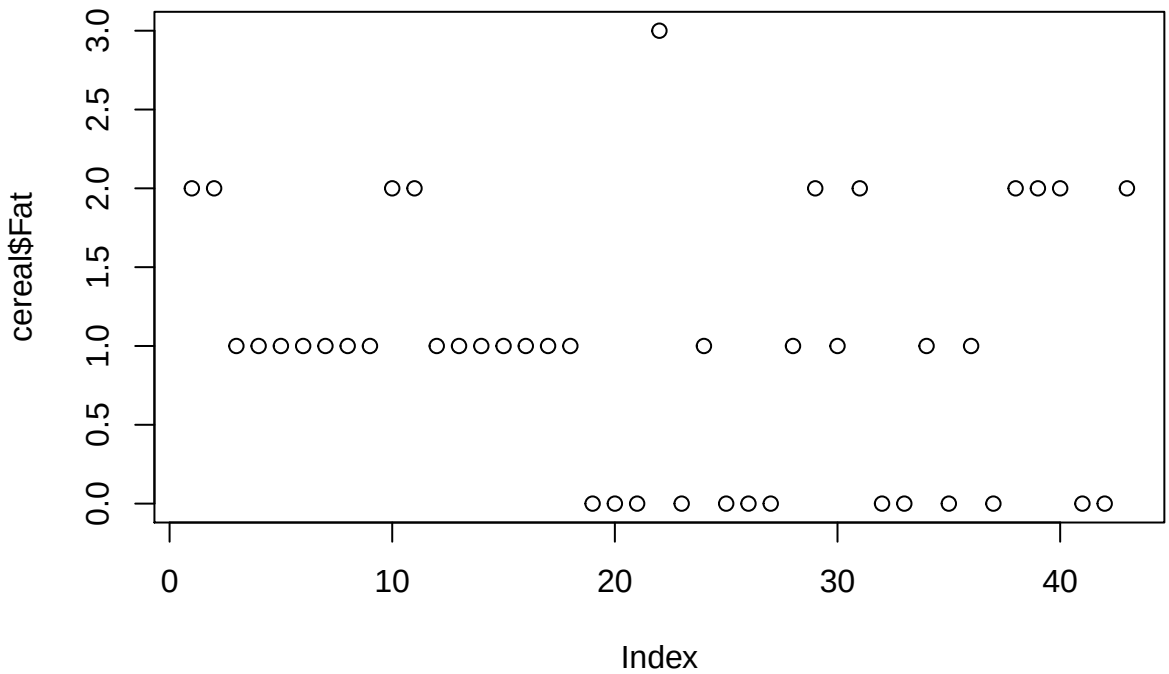


Simple one variable scatterplot

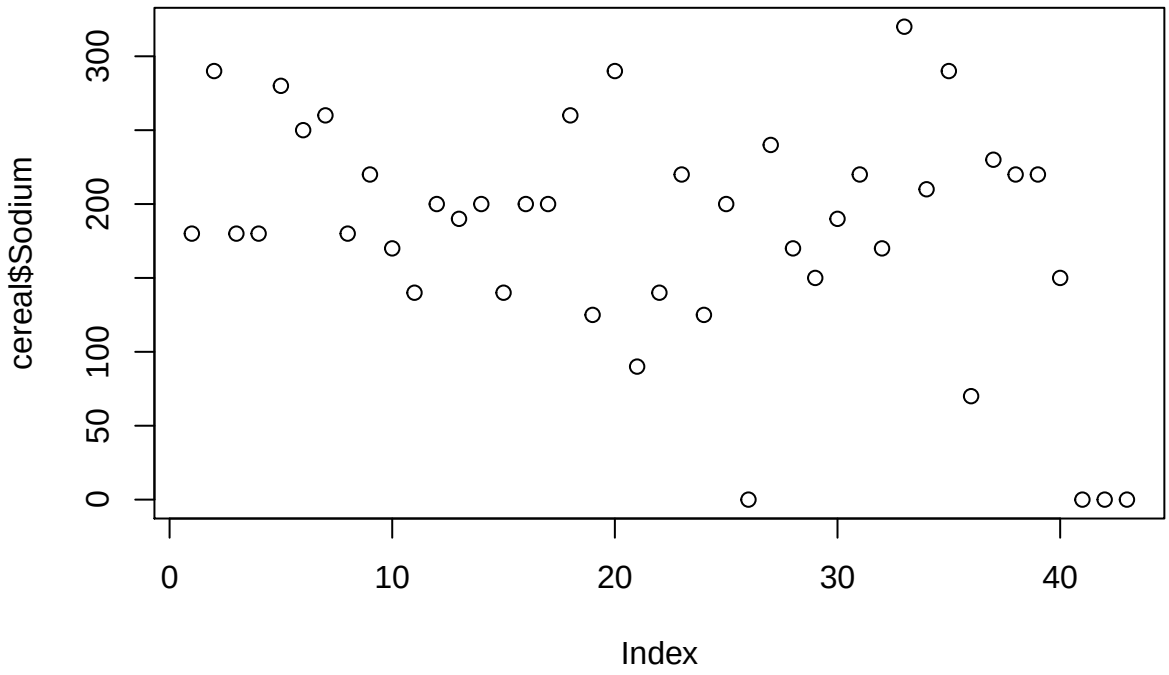
```r
plot(cereal$Calories)
```



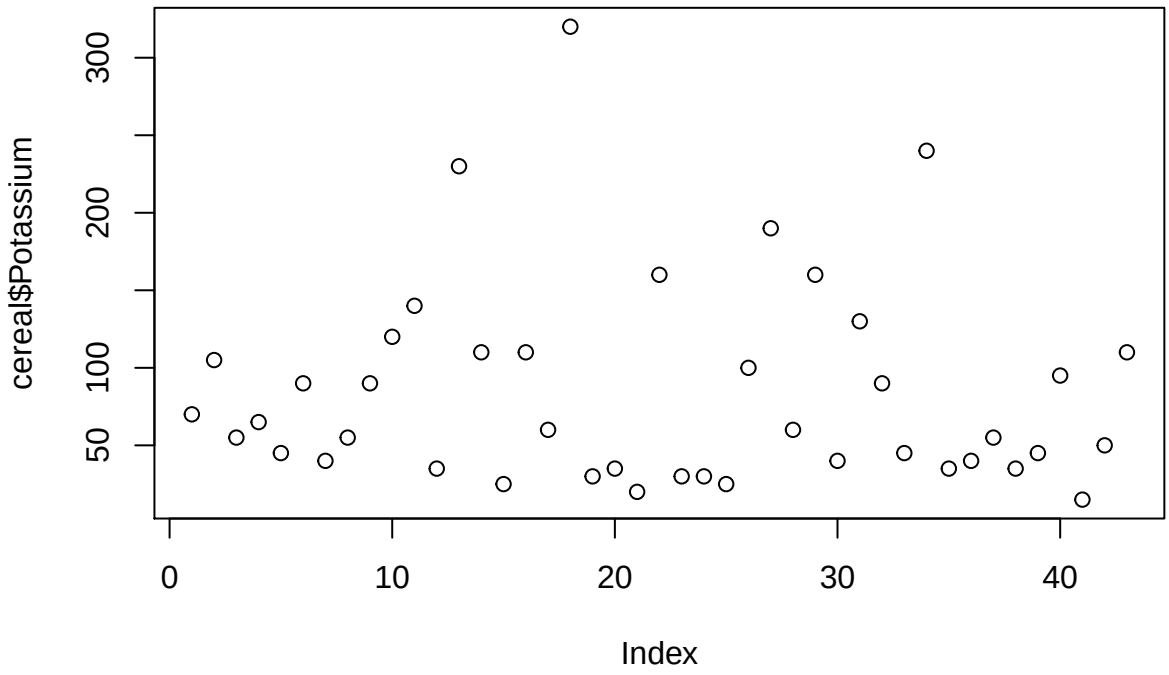
```r
plot(cereal$Fat)
```



```r
plot(cereal$Sodium)
```

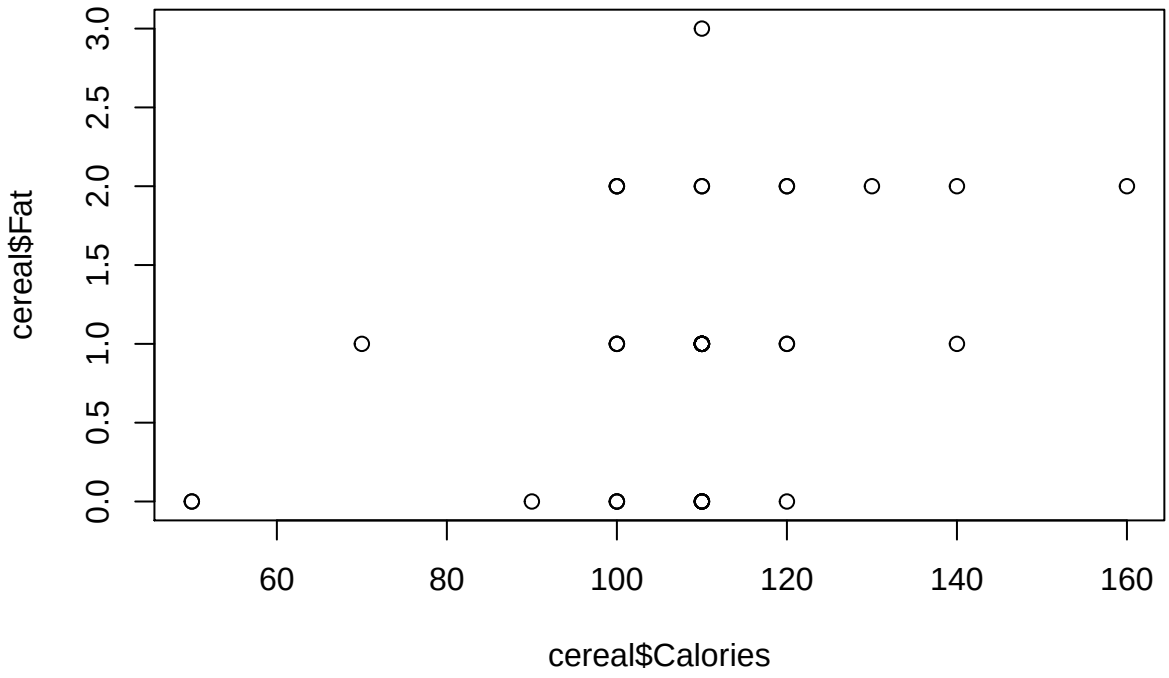


```r
plot(cereal$Potassium)
```



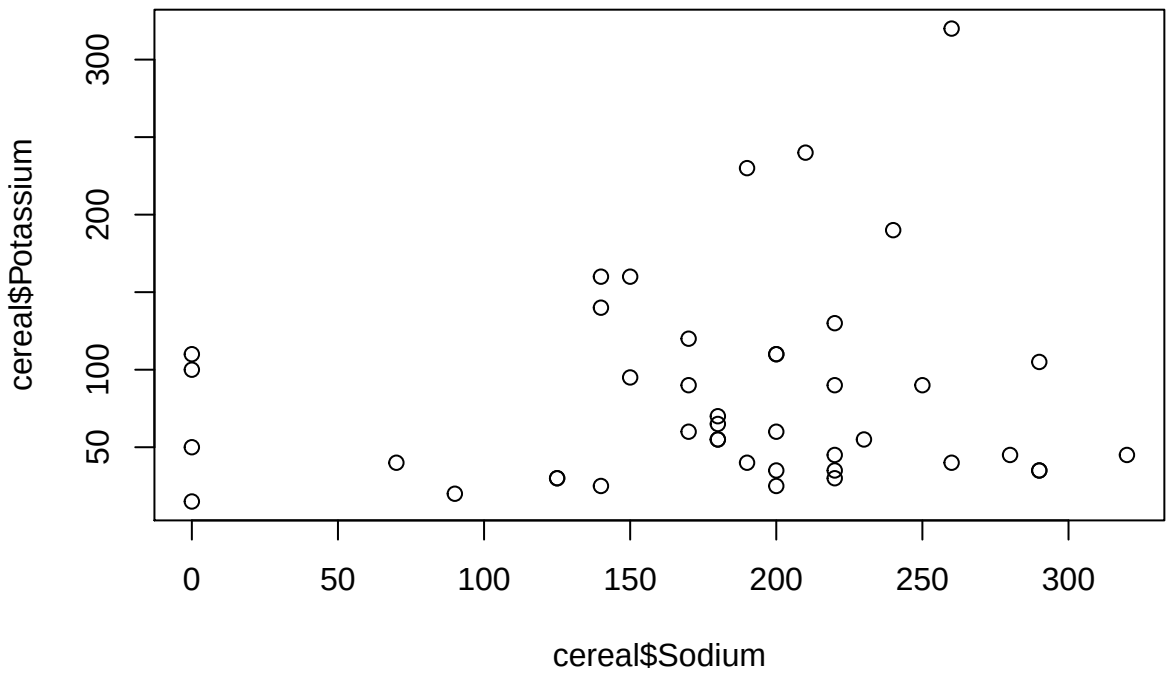
Simple two variable scatterplot

```
plot(cereal$Calories, cereal$Fat)
```



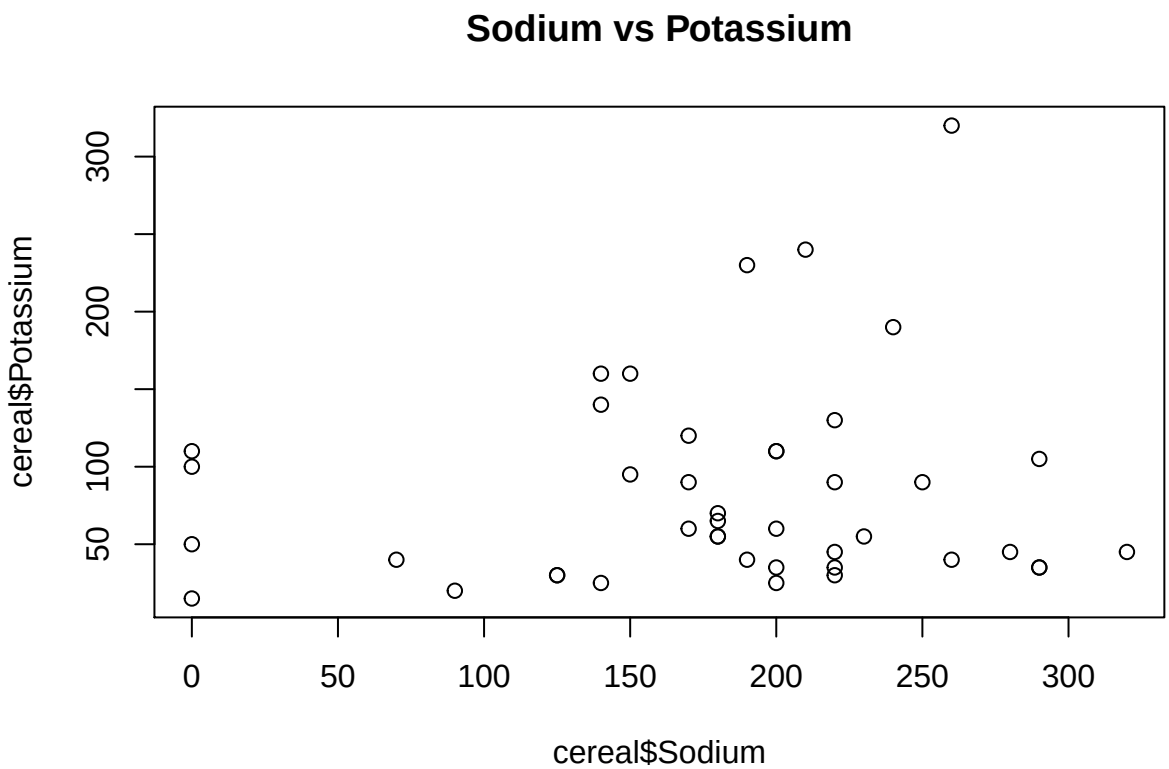
Let's look at some variables that have more possible values

```
plot(cereal$Sodium, cereal$Potassium)
```

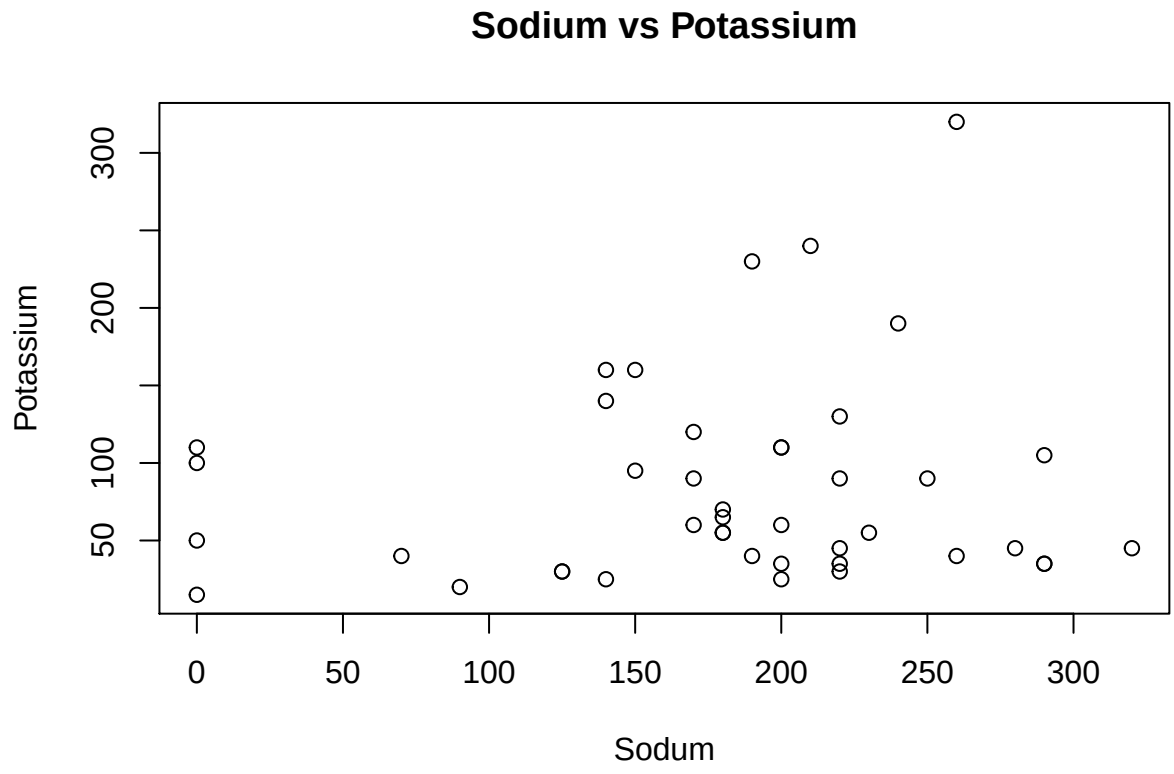


Now let's add a title

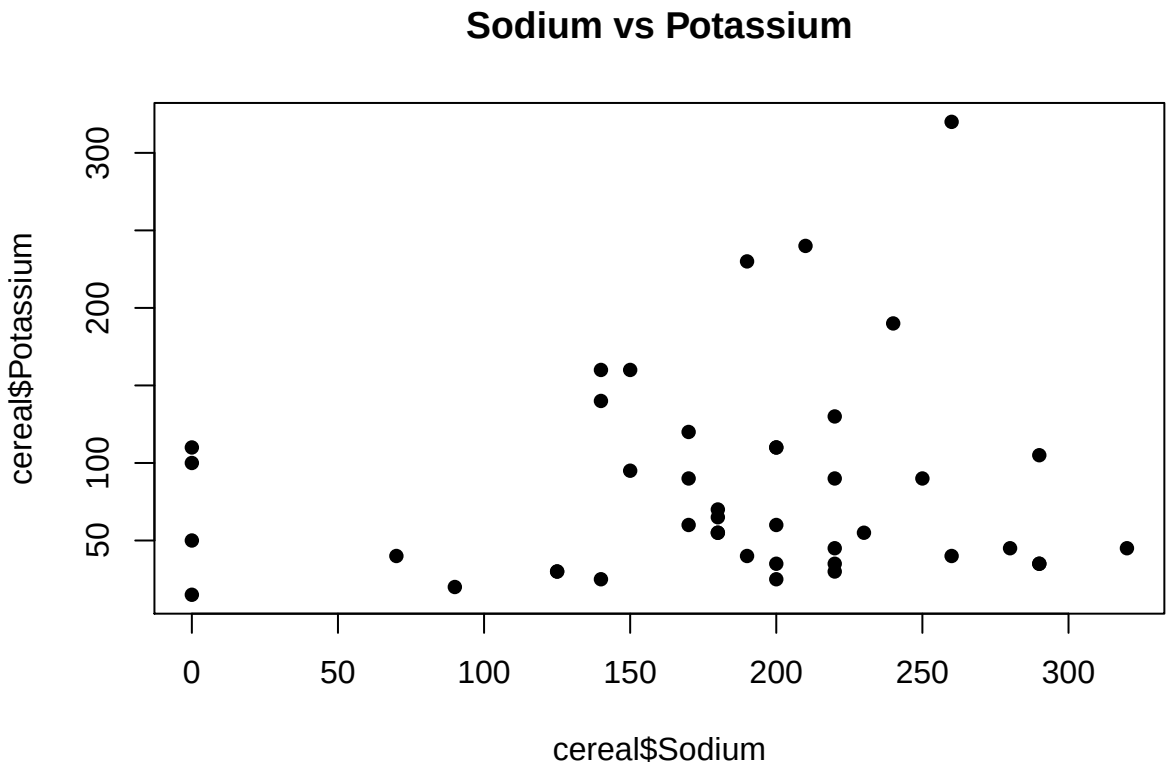
```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium')
```



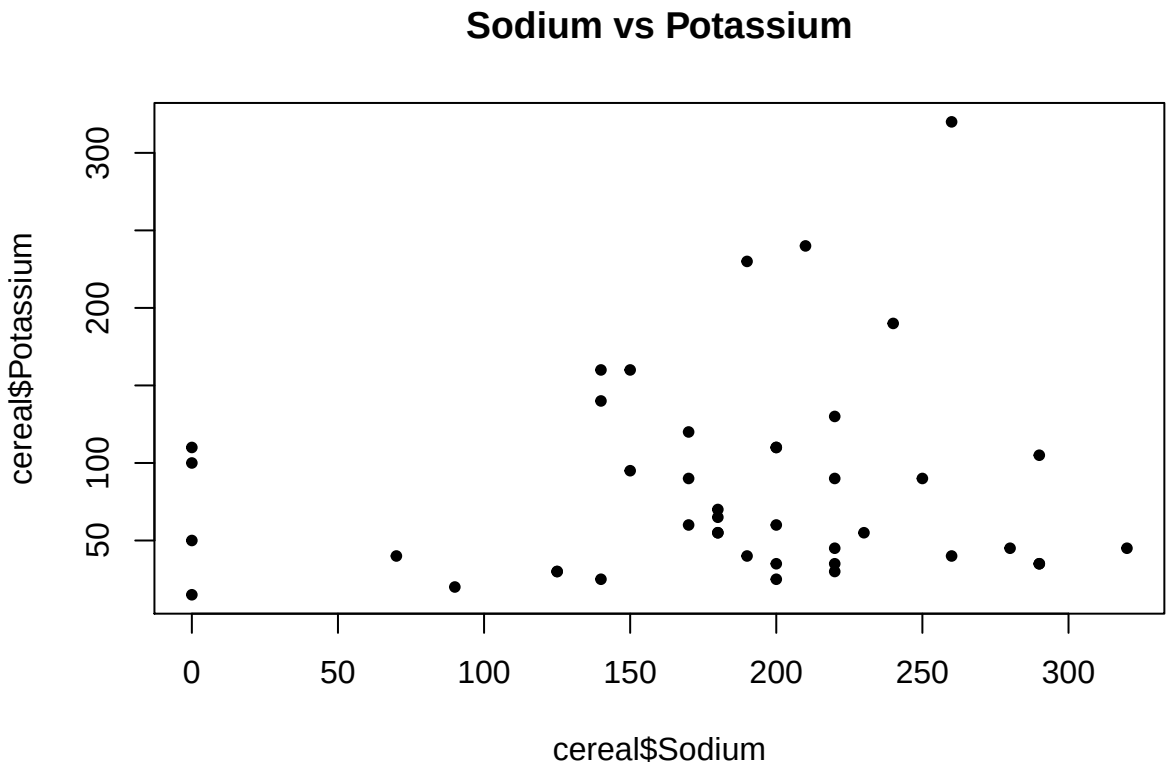
```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium', xlab = 'Sodum', ylab = 'Potassium')
```



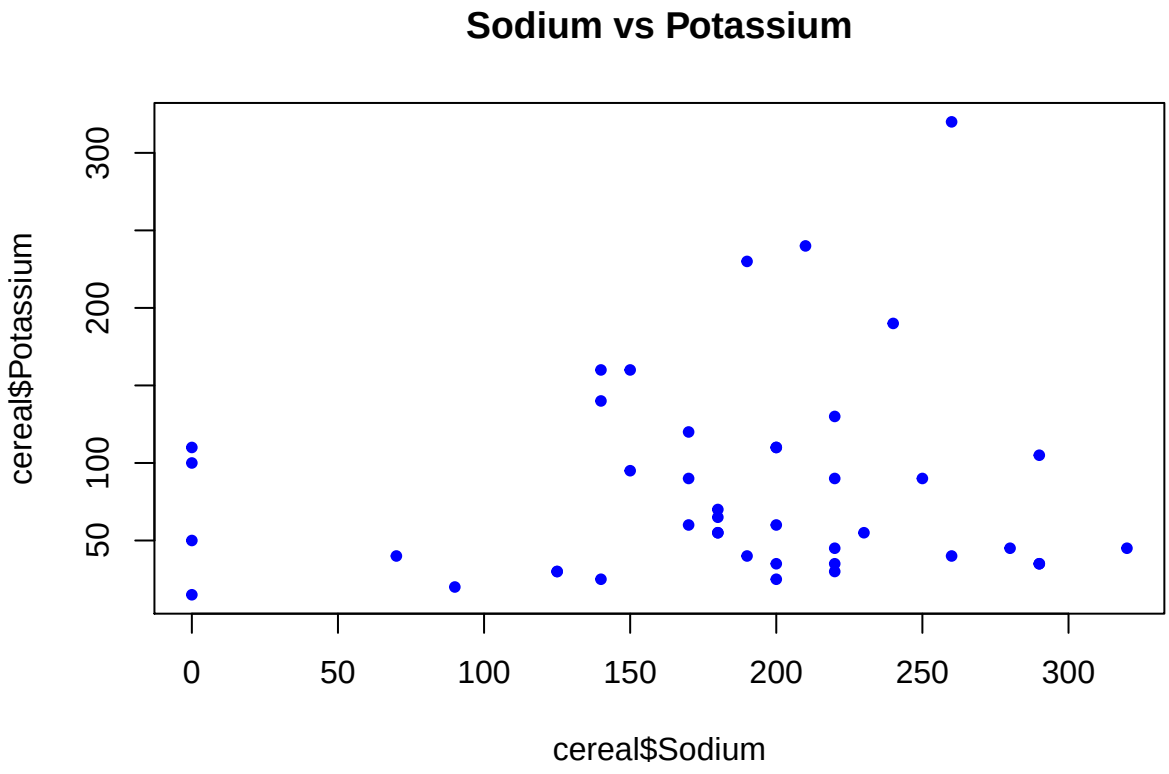
```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
```



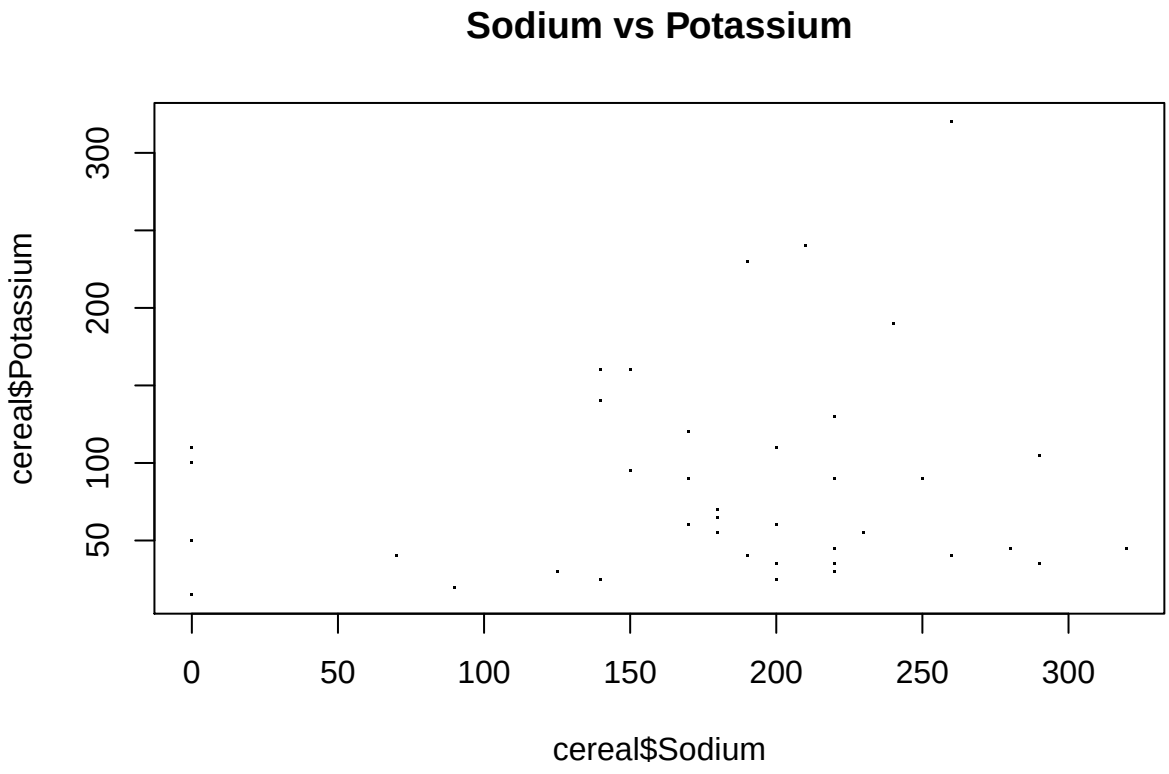
```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 20)
```



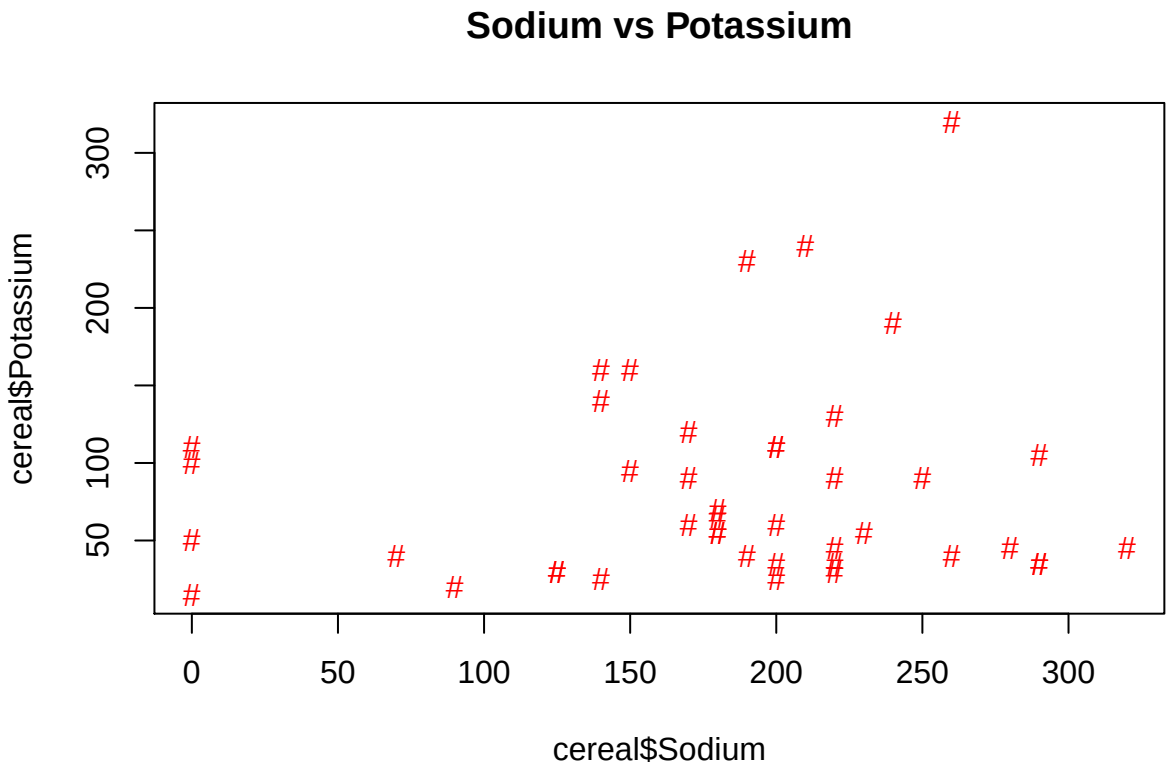
```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 20, col = 'blue')
```



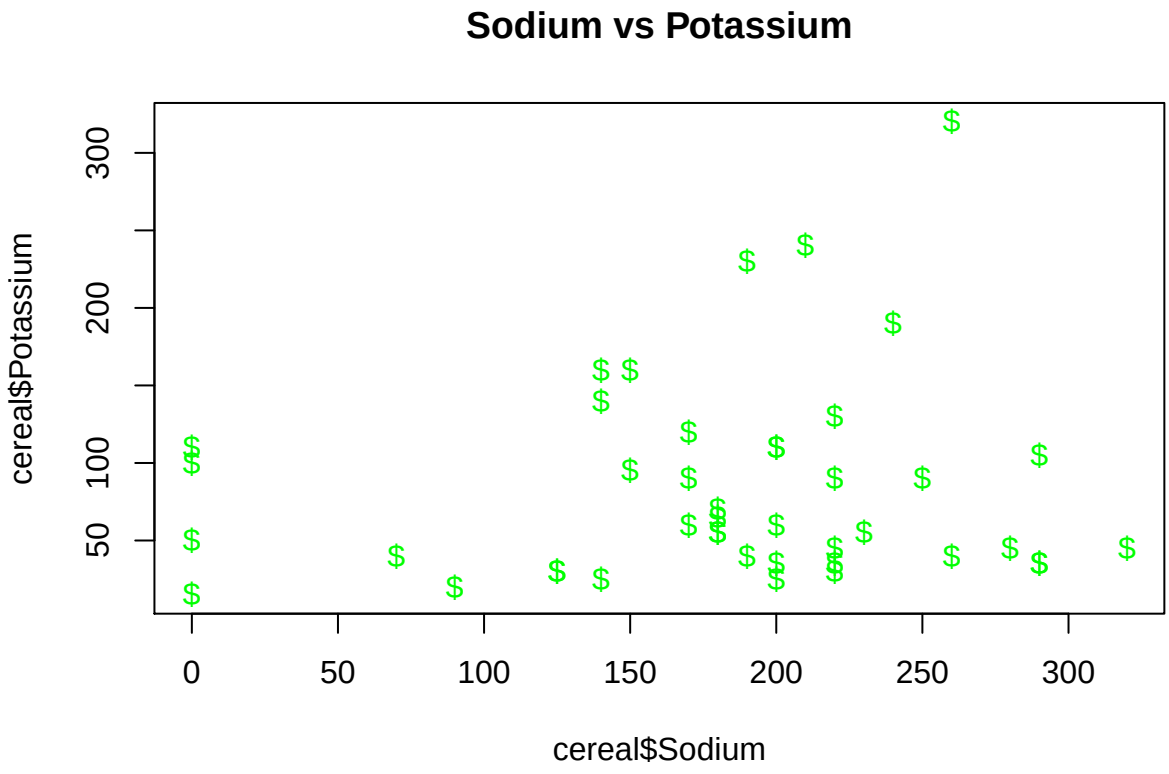
```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = '.')
```



```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = '#', col = 'red')
```

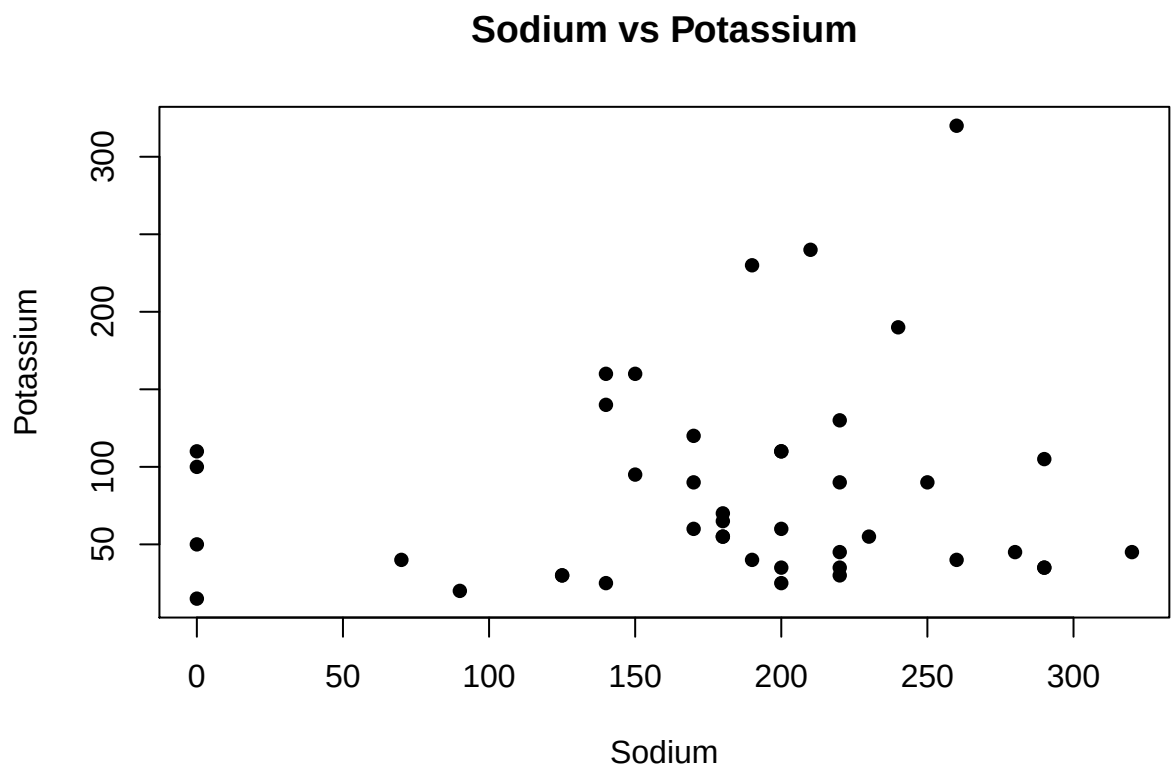


```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = '$', col = 'green')
```



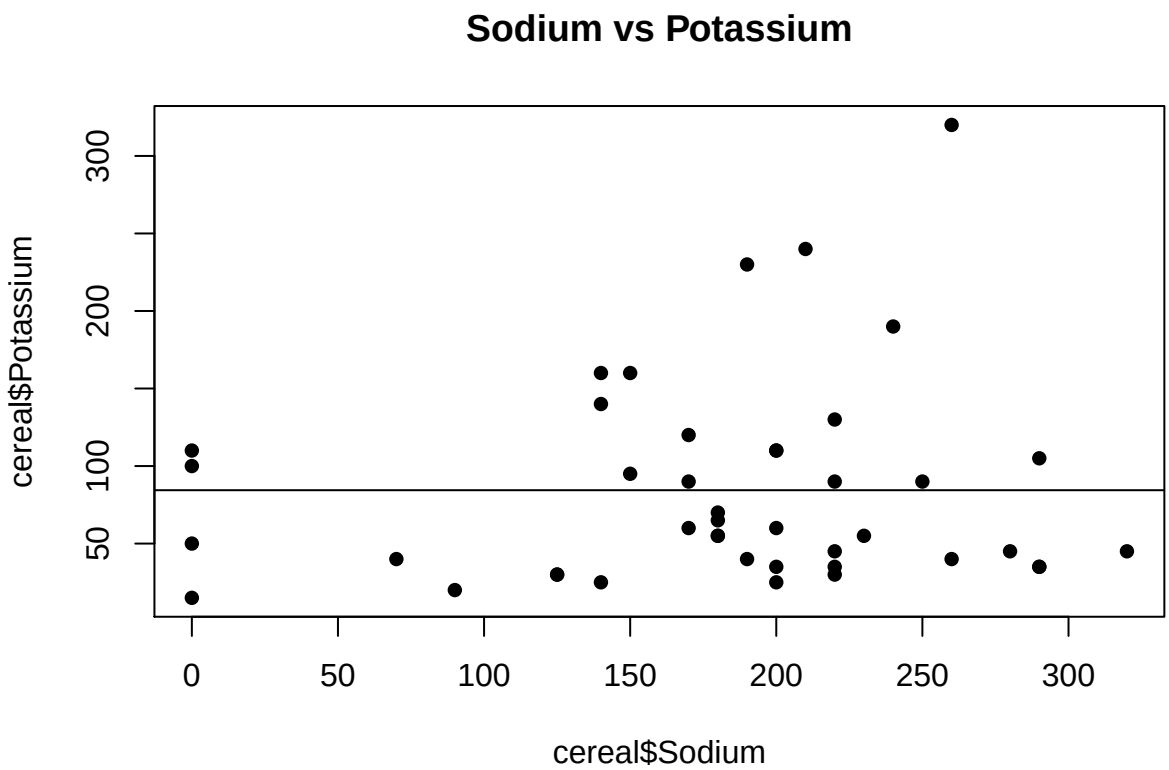
Change x and y labels

```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium', xlab = 'Sodium', ylab = 'Potassium'
```



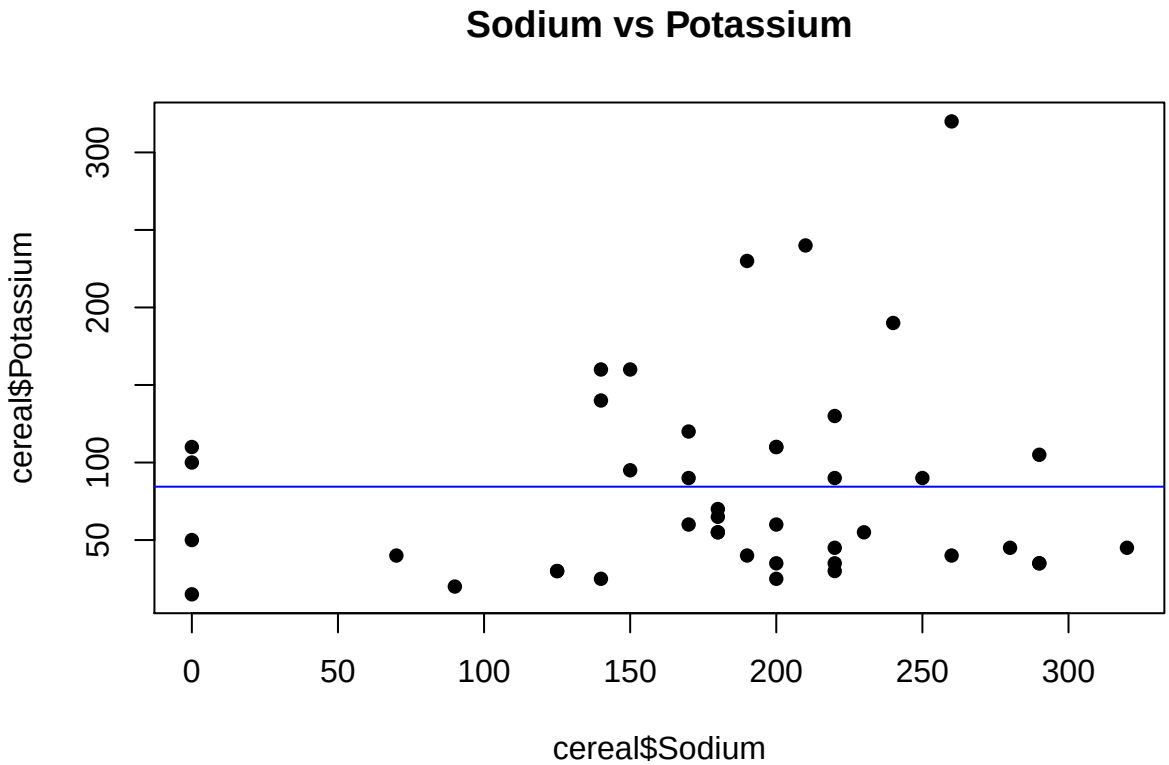
Add horizontal line to the plot

```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
abline(h = mean(cereal$Potassium))
```



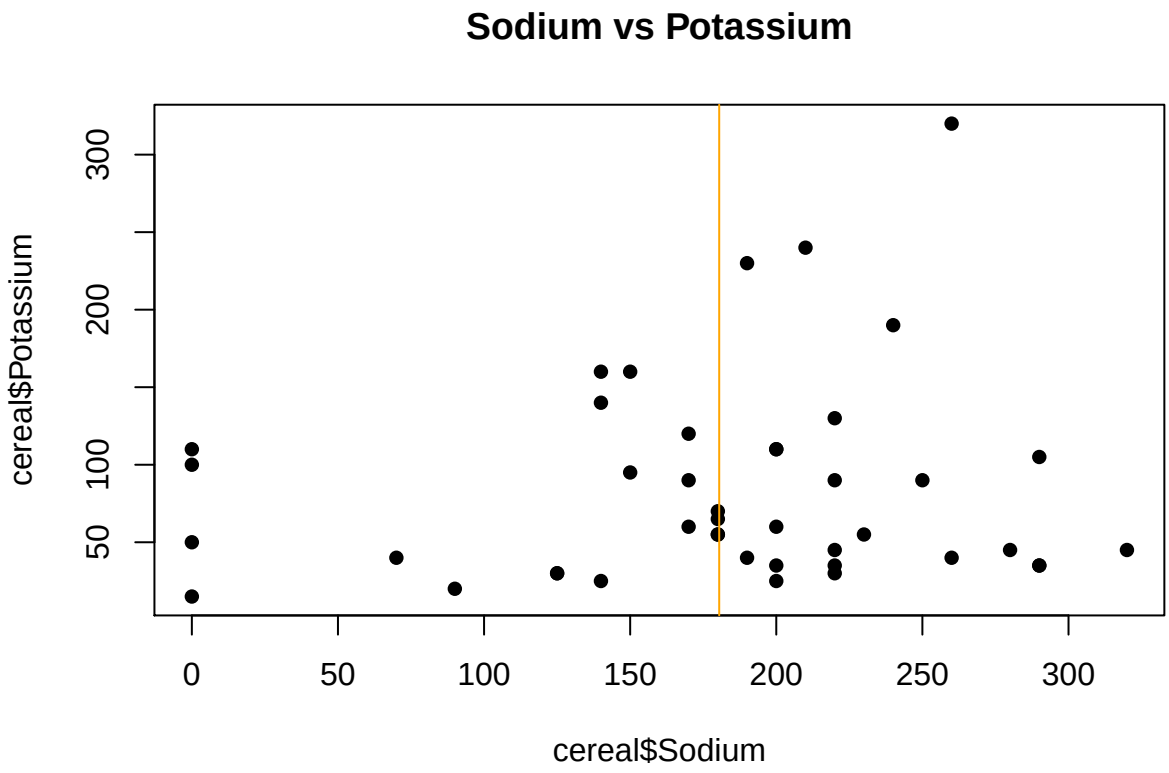
Change color of horizontal line

```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
abline(h = mean(cereal$Potassium), col = 'blue')
```



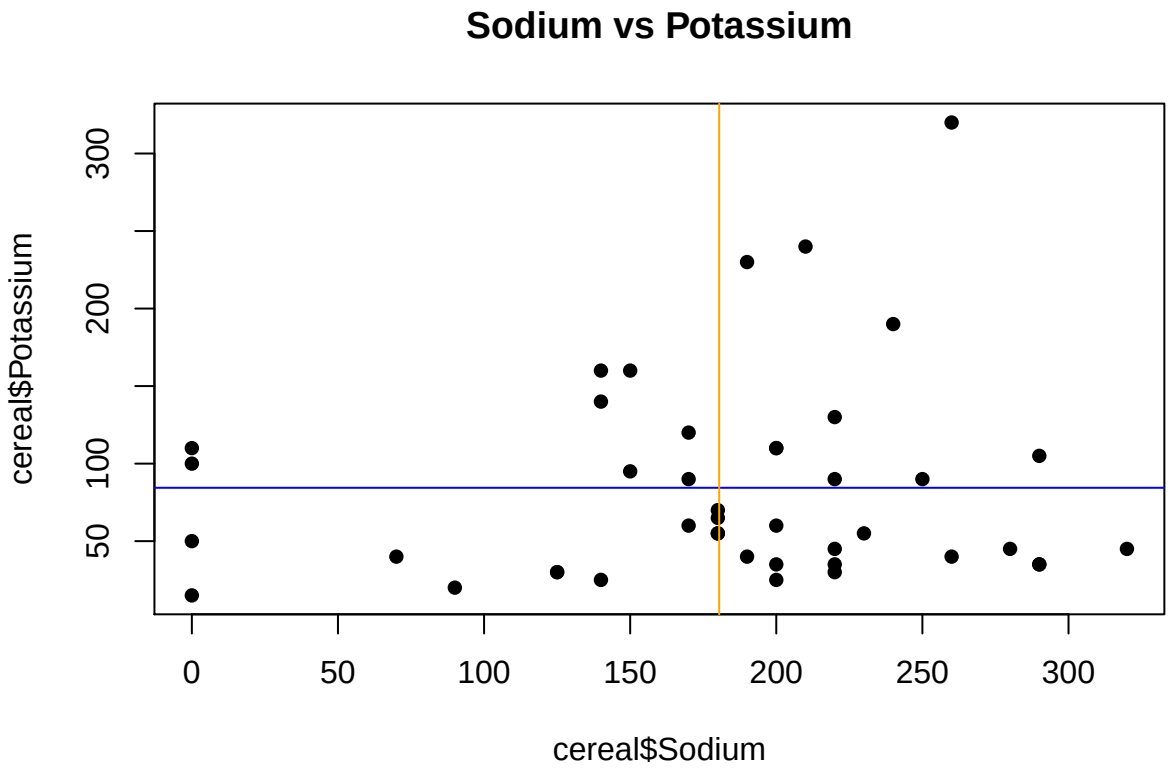
Add vertical line to plot, orange color

```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
abline(v = mean(cereal$Sodium), col = 'orange')
```



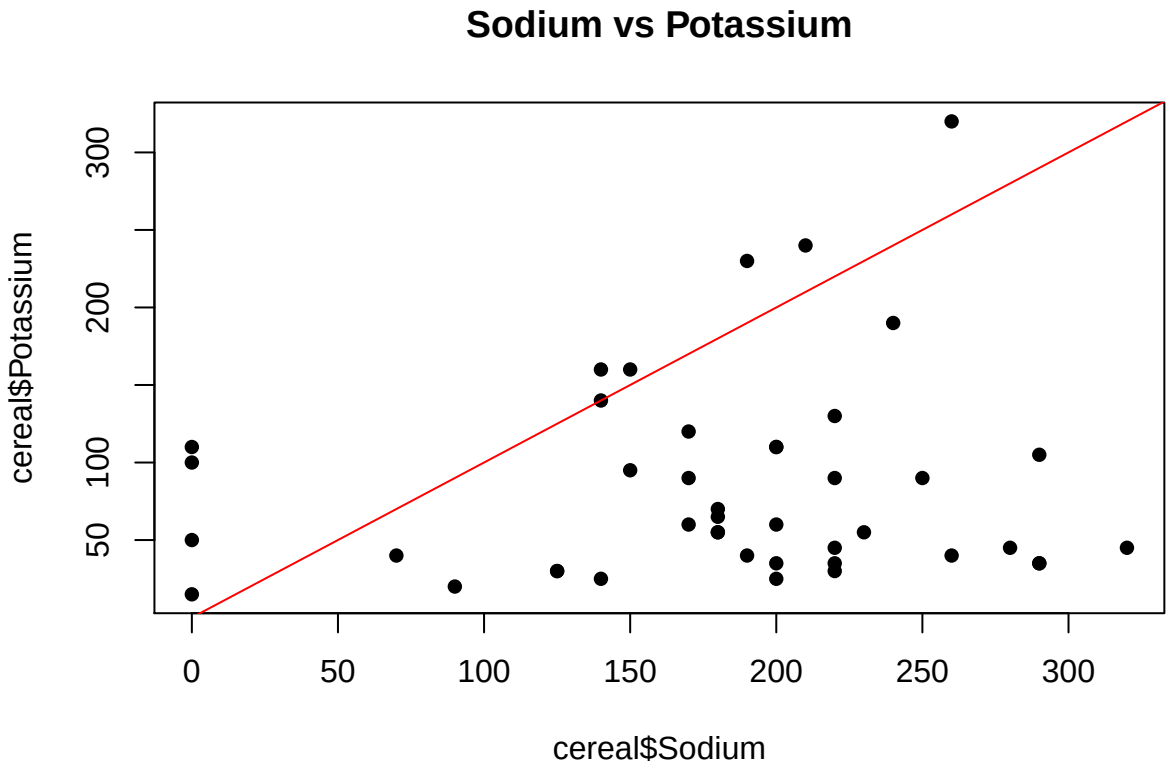
Add both vertical and horizontal lines to plot

```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
abline(h = mean(cereal$Potassium), col = 'blue')
abline(v = mean(cereal$Sodium), col = 'orange')
```



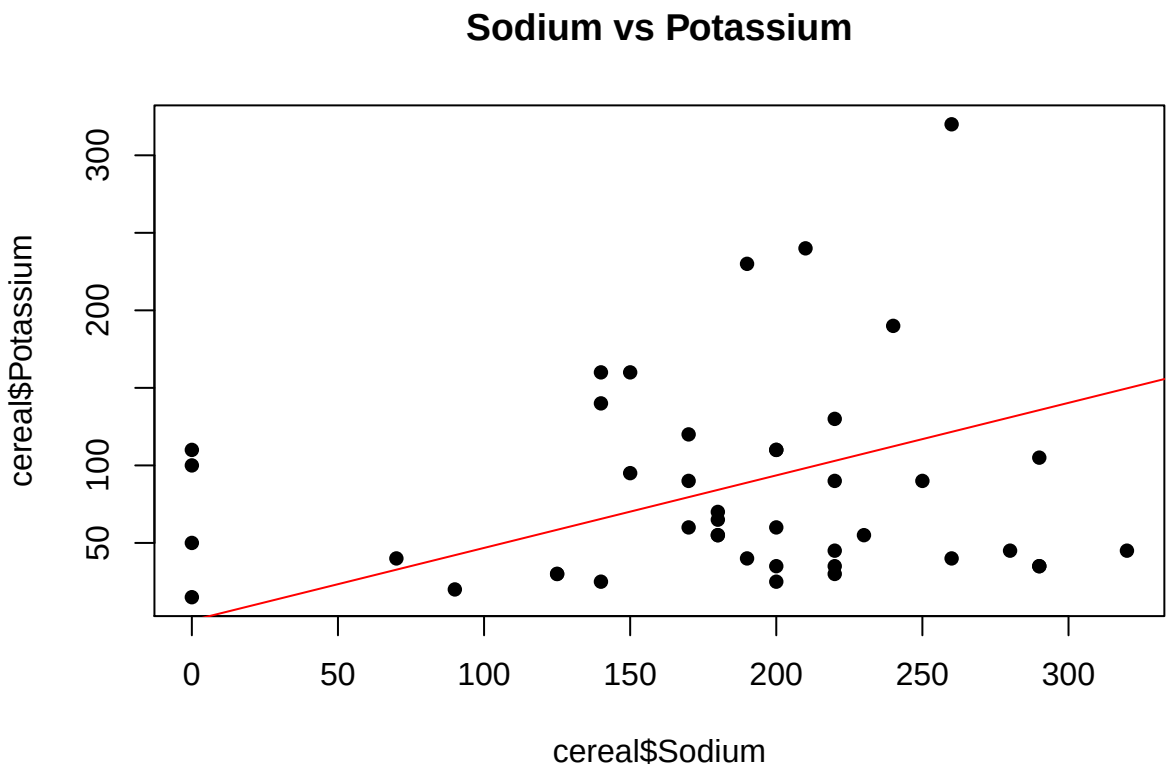
Add line with slope and intercept to plot

```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
abline(a = 0, b = 1, col = 'red')
```

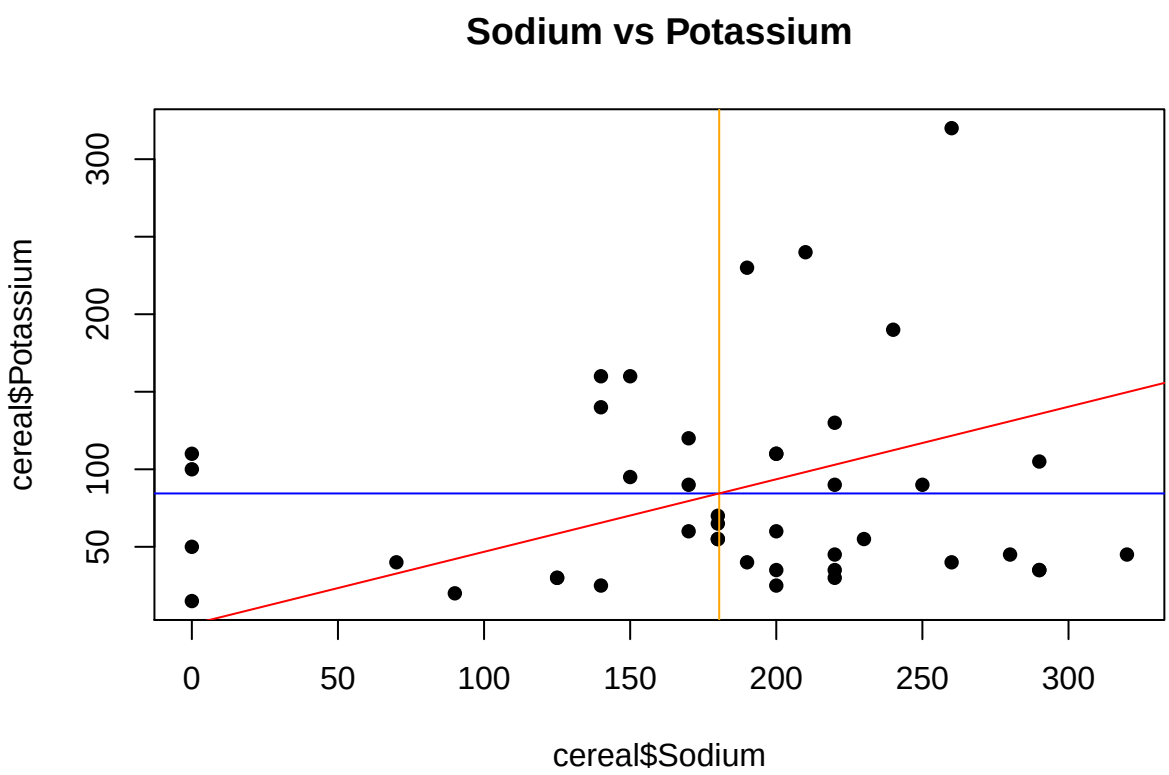


Add line with slope and intercept to plot

```
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
abline(a = 0,
       b = mean(cereal$Potassium)/mean(cereal$Sodium), col = 'red')
```



```r
plot(cereal$Sodium, cereal$Potassium, main = 'Sodium vs Potassium',pch = 16)
abline(h = mean(cereal$Potassium), col = 'blue')
abline(v = mean(cereal$Sodium), col = 'orange')
abline(a = 0,
       b = mean(cereal$Potassium)/mean(cereal$Sodium), col = 'red')
```



See https://www.statmethods.net/advgraphs/parameters.html for more coverage of graphical paramters